

CHIRAN BINNU CHERIAN

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GUIDANCE, NAVIGATION & CONTROL | SPACECRAFT ESTIMATION | ASTRODYNAMICS

EDUCATION

Iowa State University, Ames, IA	May 2026 - Present
Ph.D. in Aerospace Engineering GPA: 3.52/4.00	
Iowa State University, Ames, IA	Jan. 2024 - May 2026
M.S. in Aerospace Engineering, Thesis GPA: 3.52/4.00	
Vellore Institute of Technology, Vellore, India	Jul. 2019 - May 2023
B.Tech. in Mechanical Engineering GPA: 7.94/10.00	

TECHNICAL SKILLS

Programming: Python, MATLAB, Mathematica

Estimation & Navigation: Kalman Filtering, EKF, UKF, QEKF, QUKF, Polynomial Filtering, CUT, Monte Carlo Analysis, Orbit Determination

GNC & Astrodynamics: Stochastic Optimal Control, Covariance Steering, Nonlinear/Trajectory Optimization, Clohessy-Wiltshire Dynamics, CR3BP, Lambert Problem, Spacecraft Relative Navigation

Tools: ABAQUS, SolidWorks, Autodesk Fusion 360, OpenRocket, Google Colab, Jetstream2

GNC RESEARCH EXPERIENCE

SPACE Lab - Iowa State University May 2024 - Present

Graduate Researcher | Advisor: Dr. Simone Servadio

- Co-developed Quadratic Extended and Unscented Kalman Filters (**QEKF/QUKF**), extending EKF/UKF measurement updates from affine to **quadratic MMSE** approximations for nonlinear estimation.
- Integrated skewness and kurtosis for non-Gaussian measurement uncertainty; QUKF reduced scalar-test RMSE from **0.0395 to 0.0290 (about 27%)** and lowered position/velocity errors in **Monte Carlo spacecraft relative-navigation** tests.
- Generalized the measurement update to **arbitrary-order polynomial MMSE** estimators using Conjugate Unscented Transformation (**CUT**) rules; evaluated on Clohessy-Wiltshire and **CR3BP** dynamics with improved estimation accuracy and covariance consistency.

LEQG Covariance Steering - Iowa State University

August 2026

Researcher | Collaborators: Yasemin Isik and Dr. Abhishek Halder

- Formulated continuous-time risk-sensitive **LEQG covariance steering**, derived the optimal linear state-feedback structure, and established **local existence/uniqueness** near the risk-neutral solution.
- Implemented **Newton's method with backtracking line search** for the implicit matrix equation and demonstrated the solver on a noisy double-integrator across risk-sensitivity settings.

SELECTED GNC & AEROSPACE PROJECTS

Multi-Revolution Lambert Solver & Interplanetary Trajectory Reconstruction Aug. - Dec. 2024

- Developed a constrained **multiple-revolution Lambert solver** in MATLAB with periapsis/apoapsis restrictions; reconstructed Cassini-Huygens ephemerides, validated against **NASA/JPL Horizons**, and extended the framework to Galileo and Voyager.

Titan Parafoil Terminal-Descent Optimization

Feb. - May 2025

- Formulated a nonconvex terminal-descent optimization with glide-slope, pointing-angle, and virtual-control constraints for soft landing and **terrain-relative-navigation** requirements.

Team Sammard - University Rocket Team, VIT

Jul. 2019 - May 2023

Senior Team Member; Former Recovery Head

- Contributed to design, fabrication, launch, and recovery of a **10,000-ft sounding rocket** for the Spaceport America Cup; led recovery-system activities and improved ejection/motor-test procedures.

INDUSTRY EXPERIENCE

Dynatomic Technologies Limited - Oldland Aerospace, India

Aug. - Oct. 2023

Engineering Intern

- Supported SolidWorks-based aerospace component design/manufacturing and performed quality control, NDT, CNC machining, assembly, and test-specimen fabrication.

APT Global Marine Services L.L.C., UAE

Jun. - Aug. 2022

Engineering Intern

- Produced Fusion 360 CAD models and supported manufacturing, testing, welding, CNC machining, assembly, and quality inspection of naval components.

SELECTED PUBLICATIONS

- S. Servadio and **C. B. Cherian**, "Quadratic Extended and Unscented Kalman Filter Updates," MFI, 2025.
- C. B. Cherian**, Y. Isik, and A. Halder, "Linear Exponential Quadratic Gaussian Covariance Steering," under review; arXiv:2609.12463, 2026.
- C. B. Cherian** and S. Servadio, "Polynomial Updates for the Unscented Kalman Filter," under review; arXiv:2603.20259, 2026.